Chapter 6. Simple Linear Regression

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This chapter reviews simple linear regression, the tool behind the regression estimator of Ch. 7. Regression compresses n-dimensional data into a 2-dimensional summary $(\hat{\beta}_0, \hat{\beta}_1)$: data = model structure + error.

1 The Model

Definition 6.1

The simple linear regression model with a single regressor x is

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \qquad i = 1, \ldots, n,$$

where the errors are uncorrelated with mean 0 and unknown variance σ^2. The regressor x is treated as fixed; y is random. β_0 (intercept) and β_1 (slope) are the **regression coefficients**.

Then

$$\mathrm{E}(y \mid x) = \beta_0 + \beta_1 x = f(x), \qquad \mathrm{Var}(y \mid x) = \mathrm{Var}(\varepsilon) = \sigma^2,$$

so the true regression line is the line of mean responses. The goals are to estimate β_0, β_1 and to test $H_0 : \beta_1 = 0$.

2 Least-Squares Estimation

The least-squares (LS) estimators minimize

$$S(\beta_0,\beta_1)=\sum_{i=1}^{n}(y_i-\beta_0-\beta_1x_i)^2=\sum_{i=1}^{n}\varepsilon_i^2.$$

Setting $\partial S/\partial\beta_0=\partial S/\partial\beta_1=0$ gives the **normal equations**

$$n\hat\beta_0+\hat\beta_1\sum x_i=\sum y_i,\qquad \hat\beta_0\sum x_i+\hat\beta_1\sum x_i^2=\sum x_iy_i.$$

Theorem 6.2: LS estimators

$$\hat\beta_1=\frac{S_{xy}}{S_{xx}}=\frac{\sum(x_i-\bar x)(y_i-\bar y)}{\sum(x_i-\bar x)^2}=\frac{\sum(x_i-\bar x)y_i}{\sum(x_i-\bar x)^2},\qquad \hat\beta_0=\bar y-\hat\beta_1\bar x.$$

The **fitted values** are $\hat y_i=\hat\beta_0+\hat\beta_1x_i=\widehat{\mathrm{E}}(y_i\mid x_i)$ and the **residuals** are $e_i=y_i-\hat y_i$, so $y_i=\hat\beta_0+\hat\beta_1x_i+e_i$; residuals are the basis of model-adequacy checks (residual analysis).

Example 6.3: Rocket propellant data, $n=20$

$y=$ shear strength, $x=$ age of propellant. From $\bar x=13.3625$, $\bar y=2131.358$, $S_{xx}=1106.559$, $S_{xy}=-41112.65$:

$$\hat\beta_1=\frac{-41112.65}{1106.559}=-37.15,\qquad \hat\beta_0=2131.358+37.15\times13.3625=2627.82,$$

so $\hat y=2627.82-37.15x$. R output: intercept 2627.822 (SE 44.184), slope -37.154 (SE 2.889, $t=-12.86$, $p=1.64\times10^{-10}$).

2.1 Properties of the estimators

Theorem 6.4

$\mathrm{E}(\hat\beta_1)=\beta_1$, $\mathrm{E}(\hat\beta_0)=\beta_0$, and

$$\mathrm{Var}(\hat\beta_1)=\frac{\sigma^2}{S_{xx}},\qquad \mathrm{Var}(\hat\beta_0)=\sigma^2\Big(\frac1n+\frac{\bar x^2}{S_{xx}}\Big),\qquad \mathrm{Cov}(\hat\beta_0,\hat\beta_1)=-\bar x\frac{\sigma^2}{S_{xx}}.$$

Derivation: proof

Write $\hat\beta_1=\sum c_iy_i$ with $c_i=(x_i-\bar x)/S_{xx}$. Then $\sum c_i=0$ and $\sum c_ix_i=\sum(x_i-\bar x)x_i/S_{xx}=1$, so

$$\mathrm{E}(\hat\beta_1)=\sum c_i(\beta_0+\beta_1x_i)=\beta_0\sum c_i+\beta_1\sum c_ix_i=\beta_1,$$

$$\mathrm{Var}(\hat\beta_1)=\sigma^2\sum c_i^2=\sigma^2\frac{S_{xx}}{S_{xx}^2}=\frac{\sigma^2}{S_{xx}}.$$

Since $\mathrm{E}(\bar y)=\beta_0+\beta_1\bar x$, $\mathrm{E}(\hat\beta_0)=\beta_0+\beta_1\bar x-\bar x\beta_1=\beta_0$. With $\bar y=\sum d_iy_i$, $d_i=1/n$,

$$\mathrm{Cov}(\bar y,\hat\beta_1)=\sigma^2\sum d_ic_i=\frac{\sigma^2}{n}\sum c_i=0,$$

hence

$$\mathrm{Var}(\hat\beta_0)=\mathrm{Var}(\bar y)+\bar x^2\mathrm{Var}(\hat\beta_1)-2\bar x\mathrm{Cov}(\bar y,\hat\beta_1)=\sigma^2\Big(\frac1n+\frac{\bar x^2}{S_{xx}}\Big),$$

$$\mathrm{Cov}(\hat\beta_0,\hat\beta_1)=\mathrm{Cov}(\bar y,\hat\beta_1)-\bar x\mathrm{Var}(\hat\beta_1)=-\bar x\sigma^2/S_{xx}.$$

2.2 Properties of the fit

Theorem 6.5

(1) $\sum e_i=0$, equivalently $\sum y_i=\sum\hat y_i$;

(2) the fitted line passes through $(\bar x,\bar y)$;

(3) $\sum x_ie_i=0$;

(4) $\sum\hat y_ie_i=0$.

Proof. (1) and (3) are the two normal equations: $\sum(y_i-\hat\beta_0-\hat\beta_1x_i)=0$ and $\sum x_i(y_i-\hat\beta_0-\hat\beta_1x_i)=0$. (2): at $x=\bar x$, $\hat y=\hat\beta_0+\hat\beta_1\bar x=\bar y$. (4): $\sum\hat y_ie_i=\hat\beta_0\sum e_i+\hat\beta_1\sum x_ie_i=0$. ■

2.3 Centered model

Rewriting

$$y_i=(\beta_0+\beta_1\bar x)+\beta_1(x_i-\bar x)+\varepsilon_i=\beta_0'+\beta_1(x_i-\bar x)+\varepsilon_i$$

gives $\hat\beta_0'=\bar y$ and $\hat\beta_1=S_{xy}/S_{xx}$, and now $\mathrm{Cov}(\hat\beta_0',\hat\beta_1)=\mathrm{Cov}(\bar y,\hat\beta_1)=0$: centering makes the estimators uncorrelated.

3 Estimation of σ^2

Theorem 6.6: Sum-of-squares decomposition

$$\underbrace{\sum(y_i-\bar y)^2}_{SS_T}=\underbrace{\sum(y_i-\hat y_i)^2}_{SS_E}+\underbrace{\sum(\hat y_i-\bar y)^2}_{SS_R},\qquad SS_E=SS_T-\hat\beta_1S_{xy},\quad SS_R=\hat\beta_1S_{xy}.$$

Proof. Expanding $\sum(y_i-\hat y_i+\hat y_i-\bar y)^2$ leaves the cross term $2\sum e_i(\hat y_i-\bar y)=2(\sum e_i\hat y_i-\bar y\sum e_i)=0$. Since $e_i=(y_i-\bar y)-\hat\beta_1(x_i-\bar x)$,

$$SS_E=SS_T+\hat\beta_1^2S_{xx}-2\hat\beta_1S_{xy}=SS_T-\hat\beta_1S_{xy}.$$

■

SS_E has $n-2$ degrees of freedom, and the unbiased estimator of σ^2 is

$$\hat\sigma^2=\frac{SS_E}{n-2}=MS_E\quad\text{(the mean square error)}.$$

$\hat\sigma$ is the standard error of regression; it is a model-dependent estimate of σ.

Example 6.7: Rocket propellant

$SS_T=1{,}693{,}737.6$ and $SS_E=1{,}693{,}737.6-(-37.15)(-41112.65)=166{,}402.65$, so $\hat\sigma^2=166402.65/18=9244.59$.

4 Hypothesis Testing on the Slope

Now assume $\varepsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2)$. Then $\hat{\beta}_1 \sim N(\beta_1, \sigma^2/S_{xx})$.

Theorem 6.8: t-test for significance of regression

For $H_0 : \beta_1 = 0$ versus $H_1 : \beta_1 \neq 0$,

$$t_0 = \frac{\hat{\beta}_1}{\text{se}(\hat{\beta}_1)}, \qquad \text{se}(\hat{\beta}_1) = \sqrt{MS_E/S_{xx}},$$

which is t_{n-2} under H_0. Reject H_0 if $|t_0| > t_{\alpha/2,n-2}$ (equivalently, if the P-value $\leq \alpha$). If σ^2 were known, $Z_0 = \hat{\beta}_1/\sqrt{\sigma^2/S_{xx}} \sim N(0,1)$ would be used instead.

Failing to reject H_0 means there is no evidence of a *linear* relationship between x and y.

Example 6.9: Rocket propellant

$t_0 = -37.15/\sqrt{9244.59/1106.56} = -12.85$ and $t_{0.025,18} = 2.101$, so H_0 is rejected; the P-value is 1.64×10^{-10}.

4.1 Analysis of variance

Theorem 6.10: F-test

With degrees of freedom $(n-1) = 1 + (n-2)$,

$$F_0 = \frac{SS_R/1}{SS_E/(n-2)} = \frac{MS_R}{MS_E} \sim F_{1,n-2} \quad \text{under } H_0,$$

and H_0 is rejected if $F_0 > F_{\alpha,1,n-2}$. Since $\text{E}(MS_R) = \sigma^2 + \beta_1^2 S_{xx}$ and $\text{E}(MS_E) = \sigma^2$, F_0 is near 1 under H_0 and large otherwise. In simple regression $F_0 = t_0^2$.

ANOVA gives the same answer as the t-test here, but it is the natural tool for multiple regression.

Example 6.11: Rocket propellant

$SS_R = (-37.15)(-41112.65) = 1{,}527{,}334.95$, $SS_E = 166{,}402.65$,

$$F_0 = \frac{1527334.95/1}{166402.65/18} = 165.21 > F_{0.05,1,18} = 4.41,$$

so H_0 is rejected (R: $F = 165.4$ on 1 and 18 df, $p = 1.64 \times 10^{-10}$).

5 Interval Estimation

5.1 Coefficients

Under normal errors, $(\hat{\beta}_j - \beta_j)/\text{se}(\hat{\beta}_j) \sim t_{n-2}$, so

$$\hat{\beta}_1 \pm t_{\alpha/2,n-2}\,\text{se}(\hat{\beta}_1), \qquad \hat{\beta}_0 \pm t_{\alpha/2,n-2}\,\text{se}(\hat{\beta}_0),$$

$$\text{se}(\hat\beta_1) = \sqrt{\frac{MS_E}{S_{xx}}}, \qquad \text{se}(\hat\beta_0) = \sqrt{MS_E\Big(\frac{1}{n} + \frac{\bar x^2}{S_{xx}}\Big)}.$$

Example 6.12

For the rocket data with $t_{0.025,18} = 2.101$:

$$2534.99 \le \beta_0 \le 2720.65, \qquad -43.22 \le \beta_1 \le -31.08.$$

5.2 Error variance

Since $(n-2)MS_E/\sigma^2 \sim \chi^2_{n-2}$,

$$\frac{(n-2)MS_E}{\chi^2_{\alpha/2,n-2}} \le \sigma^2 \le \frac{(n-2)MS_E}{\chi^2_{1-\alpha/2,n-2}}.$$

5.3 Mean response

For x_0 within the range of the data, the estimate is $\hat\mu_{y|x_0} = \hat\beta_0 + \hat\beta_1 x_0$, with $\text{E}(\hat\mu_{y|x_0}) = \beta_0 + \beta_1 x_0$ and

$$\begin{aligned}\text{Var}(\hat\mu_{y|x_0}) &= \text{Var}(\hat\beta_0) + x_0^2\text{Var}(\hat\beta_1) + 2x_0\text{Cov}(\hat\beta_0,\hat\beta_1)\\ &= \sigma^2\Big(\frac{1}{n} + \frac{\bar x^2}{S_{xx}}\Big) + x_0^2\frac{\sigma^2}{S_{xx}} - 2x_0\bar x\frac{\sigma^2}{S_{xx}} = \sigma^2\Big(\frac{1}{n} + \frac{(x_0-\bar x)^2}{S_{xx}}\Big).\end{aligned}$$

Hence

$$\hat\mu_{y|x_0} \pm t_{\alpha/2,n-2}\sqrt{MS_E\Big(\frac{1}{n} + \frac{(x_0-\bar x)^2}{S_{xx}}\Big)}.$$

At $x_0 = \bar x = 13.3625$ in the rocket data the 95% interval is $2086.2 \le \text{E}(y \mid x_0) \le 2176.6$. The interval is narrowest at $\bar x$ and widens away from it.

5.4 Prediction of a new observation

The point prediction of y_0 at x_0 is $\hat y_0 = \hat\beta_0 + \hat\beta_1 x_0$. Since the new y_0 is independent of the data,

$$\text{E}(\hat y_0 - y_0)^2 = \text{E}[\hat y_0 - \text{E}(y_0)]^2 + \text{E}[y_0 - \text{E}(y_0)]^2 = \sigma^2\Big(1 + \frac{1}{n} + \frac{(x_0-\bar x)^2}{S_{xx}}\Big),$$

and the $100(1-\alpha)\%$ prediction interval is

$$\hat y_0 \pm t_{\alpha/2,n-2}\sqrt{MS_E\Big(1 + \frac{1}{n} + \frac{(x_0-\bar x)^2}{S_{xx}}\Big)}.$$

For the rocket data at $x_0 = 10$: $[2048.32,\ 2464.32]$.

6 Coefficient of Determination

Definition 6.13

$R^2 = \dfrac{SS_R}{SS_T} = 1 - \dfrac{SS_E}{SS_T}$, $0 \le R^2 \le 1$: the proportion of variation in y explained by x.

In simple regression,

$$R^2 = \frac{\hat{\beta}_1 S_{xy}}{S_{yy}} = \frac{S_{xy}^2}{S_{xx}S_{yy}} = r_{xy}^2, \qquad F_0 = \frac{SS_R/1}{SS_E/(n-2)} = \frac{R^2(n-2)}{1-R^2}.$$

For the rocket data $R^2 = 1527334.95/1693737.6 = 0.9018 = (-0.9497)^2$: 90.18% of the variability in strength is explained by age.